

CP476 Project Document

Course Registration Dashboard

Data Modeling: Entities and Relationships

Entities

- **Faculties:** Represents the university faculties.
- **Departments:** Academic departments within a faculty. Stores department name and faculty reference.
- **Programs:** Academic programs or degrees offered by departments.
- **Students:** Core student demographic and academic data, including GPA and program reference.
- **StudentAccounts and UserSessions:** Manages authentication credentials, login status, and active session tokens.
- **Instructors:** Faculty members who teach course sections.
- **Courses:** General course catalog information including course code, description, level, and credit weight.
- **AcademicTerms:** Defines specific semesters with start dates, end dates, and registration windows.
- **Rooms:** Physical campus locations with designated capacities.
- **CourseSections:** Specific offerings of a course in a given term, linking the course to an instructor, room, and schedule.
- **ClassSchedule:** Defines the specific days of the week and times a course section meets.
- **Prerequisites and Antirequisites:** Rules governing course eligibility. Prerequisites define required prior courses, while antirequisites define mutually exclusive courses.
- **Enrollments:** Resolves the many to many relationship between students and course sections, tracking registration status and final grades.
- **Waitlists:** Queues for students waiting for a seat in a full course section.

- **CoursePlans and CoursePlanItems:** Allows students to save hypothetical schedules or shopping carts prior to registration.
- **RegistrationAttempts and AuditLog:** Tracks historical registration actions, including successes, failures, and specific error reasons.
- **DegreeRequirements and RequirementCourses:** Tracks progression toward graduation by linking programs to required credit categories and specific courses.
- **Holds:** Administrative blocks (financial or academic) that prevent a student from registering.
- **SavedFilters:** Persisted search preferences for the course catalog.

Relationships

- A faculty contains multiple departments.
- A department offers multiple programs, owns multiple courses, and employs multiple instructors.
- A program enrolls multiple students and has multiple degree requirements.
- A student has one account, which can have multiple active sessions.
- A student can have multiple holds, saved filters, course plans, and entries in the audit log.
- A course has multiple sections and can have multiple prerequisites or antirequisites.
- An academic term contains multiple course sections.
- A room hosts multiple course sections.
- An instructor teaches multiple course sections.
- A course section has multiple class schedule entries.
- Students and course sections have a many to many relationship, resolved by the enrollments and waitlists tables.
- Course plans and course sections have a many to many relationship, resolved by the course plan items table.
- Degree requirements and courses have a many to many relationship, resolved by the requirement courses table.